

Zekun Wang

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Research Interests

My research focuses on **continual learning of generative foundation models** such as LLMs and diffusion models, with the goal of building agents whose behaviors (planning, reasoning, control) keep evolving through interaction instead of freezing at deployment. I study three threads: the **geometry of forgetting** (Fisher information, trust regions, rank-1 approximations), **self-improving algorithms** (meta-RL self-consolidation, curvature-aware generative replay, compositional test-time inference), and **concept discovery** as a substrate for transferring new skills onto an agent’s existing ones. I have applied these principles in three settings: a robotic policy in Trust-Region CL, a large language model in SCoL, and a diffusion model in Rank-1 Fisher and Test-Time Composition.

Education

Georgia Institute of Technology <i>Ph.D. in Computer Science</i> Advisor: Christopher J. MacLellan	2024 – 2029
University of Michigan <i>M.S. in Computer Science</i>	2022 – 2024
Pennsylvania State University <i>B.S. in Computer Science</i> Double major: Mathematics. Minor: Economics.	2018 – 2022

Selected Publications and Projects

Self-Consolidating Language Models: Continual Knowledge *In Submission*

Incorporation from Context

Zekun Wang*, Anant Gupta, Zihan Dong, Christopher J. MacLellan

- Built **SCoL**, a meta-reinforcement-learning post-training pipeline that teaches an LLM to emit *textual update directives* selecting which Transformer layers to LoRA-adapt from streaming context, balancing acquisition against forgetting.
- Beat prompting, summarization, batch test-time training, and sequential fine-tuning on SQuAD QA and LongBench v2; the learned layer selections concentrate on high-Fisher-information layers, recovering the rank-1 forgetting geometry from our prior diffusion work.

Test-Time Compositional Generalization in Diffusion Models via

In Submission

Concept Discovery

Zekun Wang*, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan

- Recover *query-specific concepts* from a pretrained diffusion model’s time-indexed scores via multi-noise-level gradient ascent, map modes to clean-space Gaussians, and compose them into a product-of-experts teacher with an analytic score.
- PoE sampler and a low-rank distilled variant beat query-only and nearest-class baselines on ColorMNIST and CelebA composition — enabling out-of-distribution composition *without retraining* the base diffusion model.

Trust Region Continual Learning as an Implicit Meta-Learner

In Submission

Zekun Wang*, Anant Gupta, Christopher J. MacLellan

- Combined generative replay with an EWC Fisher-metric trust region and *proved* that, under local approximations, the update collapses into a one-step MAML form where replay acts as the query gradient and the Fisher as offline curvature.

- On ImageNet-500 diffusion generation and Continual-World-10 diffusion policies, beats EWC, replay, FTML, and VR-MCL on both retention and re-convergence speed — bringing meta-learning theory to bear on continual robotic control.

Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion

ICLR 2026

Models

Zekun Wang*, Anant Gupta*, Zihan Dong, Christopher J. MacLellan

- Showed both mathematically and empirically that diffusion gradients become **rank-1** and aligned with the mean gradient at low SNR; built a rank-1 EWC penalty paired with generative replay.
- Forgetting is **halved or nearly eliminated** on class-incremental MNIST, FashionMNIST, CIFAR-10, and ImageNet-1k versus EWC and replay baselines.

Deep Taxonomic Networks for Unsupervised Hierarchical Prototype

NeurIPS 2025

Discovery

Zekun Wang, Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan

- Proposed deep taxonomic networks: a VAE whose prior is a complete binary tree of Gaussian mixtures, optionally combined with contrastive learning — no prior knowledge of class count required.
- Achieves **SOTA on hierarchical prototype discovery** under an evaluation that scores prototypes at every tree level; proved that the ELBO also optimizes mutual information between latent and tree.

All Publications

Test-Time Compositional Generalization in Diffusion Models via

May 2026

Concept Discovery

Zekun Wang, Anant Gupta, Tianyi Zhu, Christopher J. MacLellan

arXiv preprint

Self-Consolidating Language Models: Continual Knowledge

May 2026

Incorporation from Context

Zekun Wang*, Anant Gupta, Zihan Dong, Christopher J. MacLellan

arXiv preprint

Probabilistic Concept Formation for Lifelong and Hierarchical Topic

Apr. 2026

Modeling

Karthik Singaravadivelan*, Anant Gupta*, **Zekun Wang**, Christopher J. MacLellan

ACL 2026 (63rd Annual Meeting of the Association for Computational Linguistics)

A Rational Account of Categorization Based on Information Theory

Mar. 2026

Christopher J. MacLellan, Karthik Singaravadivelan, Xin Lian, **Zekun Wang**, Pat Langley

CogSci 2026 (48th Annual Meeting of the Cognitive Science Society)

Trust Region Continual Learning as an Implicit Meta-Learner

Feb. 2026

Zekun Wang*, Anant Gupta, Christopher J. MacLellan

arXiv preprint

Grounded Concreteness: Human-Like Concreteness Sensitivity in

Jan. 2026

Vision-Language Models

Aryan Roy, **Zekun Wang**, Christopher J. MacLellan

ACL 2026 (63rd Annual Meeting of the Association for Computational Linguistics)

Computer Vision Modeling of the Development of Geometric and

Nov. 2025

Numerical Concepts in Humans

Zekun Wang, Sashank Varma

AAAI 2026 (40th Annual AAAI Conference on Artificial Intelligence)

Hierarchical Semantic Retrieval with Cobweb

Oct. 2025

Anant Gupta*, Karthik Singaravadivelan*, **Zekun Wang**

ACS 2025 (12th Annual Conference on Advances in Cognitive Systems)

- Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models** *Sept. 2025*
Zekun Wang*, Anant Gupta*, Zihan Dong, Christopher J. MacLellan
ICLR 2026 (14th International Conference on Learning Representations)
- Deep Taxonomic Networks for Unsupervised Hierarchical Prototype Discovery** *Sept. 2025*
Zekun Wang, Ethan Haarer, Tianyi Zhu, Zhiyi Dai, Christopher J. MacLellan
NeurIPS 2025 (39th Conference on Neural Information Processing Systems)
- Efficient and Scalable Masked Word Prediction Using Concept Formation** *Jun. 2025*
 Xin Lian*, **Zekun Wang***, Christopher J. MacLellan
Cognitive Systems Research
- Taxonomic Networks: A Representation for Neuro-Symbolic Pairing** *May 2025*
Zekun Wang, Ethan Haarer, Nicki Barari, Christopher J. MacLellan
NeuS 2025 (2nd International Conference on Neuro-symbolic Systems)
- Computer Vision Models Show Human-Like Sensitivity to Geometric and Topological Concepts** *May 2025*
Zekun Wang, Sashank Varma
CogSci 2026 (48th Annual Meeting of the Cognitive Science Society)
- Babysit a Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations** *Jan. 2025*
 Ziqiao Ma*, **Zekun Wang***, Joyce Chai
NAACL 2025 (Annual Conference of the Nations of the Americas Chapter of the ACL)
- Has It All Been Solved? Open NLP Research Questions Not Solved by Large Language Models** *May 2023*
 Oana Ignat, Zhijing Jin, Artem Abzaliev, Laura Biester, Santiago Castro, Naihao Deng, Xinyi Gao, Aylin Gunal, Jacky He, Ashkan Kazemi, Muhammad Khalifa, Namho Koh, Andrew Lee, Siyang Liu, Do June Min, Shinka Mori, Joan Nwatu, Veronica Perez-Rosas, Siqi Shen, **Zekun Wang**, Winston Wu, Rada Mihalcea
LREC-COLING 2024 (Joint International Conference on Computational Linguistics, Language Resources and Evaluation)
- Foundations for NLP-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes** *May 2022*
 Susan E. Lloyd, Matthew D. Beckman, Dennis K. Pearl, Rebecca J. Passonneau, Zhaohui Li, **Zekun Wang**
ICOTS 11 (11th International Conference on Teaching Statistics)

Research Experiences

- Georgia Institute of Technology** *Aug. 2024 – present*
 Graduate Research Assistant — Teachable AI Lab
- University of Michigan** *2022 – 2024*
 Graduate Research Assistant — SLED Lab (PI: Joyce Chai)
- Pennsylvania State University** *2021 – 2022*
 Undergraduate Research Assistant

Teaching Experiences

Graduate Student Instructor — EECS 595, Natural Language Processing <i>University of Michigan</i>	<i>Sept. 2023 – May 2024</i>
Teaching Assistant — CMPSC 464, Introduction to the Theory of Computation <i>Pennsylvania State University</i>	<i>Jan. 2022 – May 2022</i>
Teaching Assistant — MATH 230, Calculus and Vector Analysis <i>Pennsylvania State University</i>	<i>Aug. 2021 – Dec. 2021</i>
Math Peer Tutor — Calculus <i>Penn State Learning</i>	<i>Jan. 2020 – Jan. 2022</i>

Invited Talks and Guest Lectures

Intro to AI (CS 3600), Georgia Tech — Guest lecture	<i>Aug. 2025</i>
Deep Learning: Classics and Trends (DLCT) — Invited talk <i>Babysit a Language Model From Scratch: Interactive Language Learning by Trials and Demonstrations</i>	<i>July 2024</i>
Advanced Artificial Intelligence (EECS 692), U. of Michigan — Guest lecture	<i>Jan. 2023</i>

Academic Service

Student Volunteer: ACS.
Conference Reviewing: NeurIPS, ICML, ICLR, AAAI, ACL, NAACL, ACS.

Academic Advising

Anant Gupta: M.S., Georgia Tech. 2025 – present.
Karthik Singaravadivelan: B.S., Georgia Tech. 2025 – present.
Aryan Roy: M.S., Georgia Tech. 2025 – present.
Zihan Dong: M.S., Georgia Tech. / Lenovo. 2024 – present.
Ethan Haarer: M.S., Georgia Tech. 2024 – 2025.

Skills

Programming	Python, C/C++, CUDA, R, MATLAB, Java, C#, TeX, SQL
ML Frameworks	PyTorch, Hugging Face (Transformers, Diffusers, PEFT/LoRA, TRL), Accelerate, DeepSpeed, vLLM, JAX (working), Weights & Biases
Infrastructure	SLURM, Docker, multi-node A100/H100 clusters, Linux/Bash, Git
Mathematics	Information / probability theory, Bayesian inference, calculus, linear algebra, variational methods, score matching, optimization
Research areas	Continual / lifelong learning, LLM post-training (RLHF/PPO/DPO/meta-RL), diffusion & flow models, meta-learning, compositional generalization, concept formation
Languages	English (fluent), Cantonese (fluent), Mandarin Chinese (native)